Faculty of Economics and Business Administration
Business Informatics and Operations Management

Supply Chain Management
Group assignment

prof. dr. LP Kerkhove

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GROUP ASSIGNMENT

1. Groups should now be available to create on Ufora, please create groups of 5 students.
2. Your solution will be evaluated on both quantitative and qualitative aspects; there is no single optimal solution – but some solutions are definitely better than others.

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SUSTAINABLE ESSENTIALS CONTEXT

1. Sustainability-Driven Business Model
2. Supply Chain Transparency
3. Circular Economy Initiatives
4. Localized and Efficient Production
5. Operational Challenges

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THE PRE-PACK CONCEPT

- Decrease packing material use and the handling required in the central distribution center
- Products are bundled at the facility where they are produced, and then shipped

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SOURCE MATERIALS

Case study (PDF)

SUPPLY CHAIN MANAGEMENT GROUP ASSIGNMENT

Your objective is to analyze and solve the pre-pack problem for Sustainable Essentials – an apparel retail company. The desired result is a technical solution, as well as a critical evaluation of the limitations of your approach within the context of the problem.

EXPECTED DELIVERABLES

I. REPORT

A report containing the details of your analysis. This report should at least cover the following elements:

- **Strategic analysis (max 1 page)**: Analyze the context and the nature of the problem. Indicate why this has led you to make specific choices in your solution methods. Here you can also refer to aspects that are outside of the scope of your quantitative models but are also relevant in this context.
- **Forecasting methodology**: Explain why you have selected specific forecasting techniques, and how you have evaluated (compared) your results. Provide information on the evaluation metrics you have used.
- **Optimization methodology**: Explain how you have calculated the optimal content of packs, and the way in which these packs are to be allocated to the different sales channels. Provide information on the tooling (software) you have used, as well as a mathematical formulation of the problem you have optimized.

3 data spreadsheets

	A	B	C	D	E
1	product_id	channel_id	season	size	units
2	IMFA21HL\	Stockholm	2019	XS	15
3	IMFA21HL\	Stockholm	2019	S	37
4	IMFA21HL\	Stockholm	2019	M	33
5	IMFA21HL\	Stockholm	2019	L	43
6	IMFA21HL\	Stockholm	2019	XL	15
7	IMFA21HL\	Stockholm	2019	XXL	15
8	IMFA21HL\	Stockholm	2020	XS	13
9	IMFA21HL\	Stockholm	2020	S	31
10	IMFA21HL\	Stockholm	2020	M	30
11	IMFA21HL\	Stockholm	2020	L	50
12	IMFA21HL\	Stockholm	2020	XL	9
13	IMFA21HL\	Stockholm	2020	XXL	12

DELIVERABLES

1/4: Report

- **Strategic analysis (max 1 page)**: Analyze the context and the nature of the problem. Indicate why this has led you to make specific choices in your solution methods. Here you can also refer to aspects that are outside of the scope of your quantitative models but are also relevant in this context.

- **Forecasting methodology**: Explain why you have selected specific forecasting techniques, and how you have evaluated (compared) your results. Provide information on the evaluation metrics you have used.

- **Optimization methodology**: Explain how you have calculated the optimal content of packs, and the way in which these packs are to be allocated to the different sales channels. Provide information on the tooling (software) you have used, as well as a mathematical formulation of the problem you have optimized.

- **Quantitative conclusions**: Briefly summarize your quantitative results, and their implications for the operational processes. Highlight cost reductions compared to the current situation.

- **Strategic conclusions decisions**: Discuss the implications of these results on a strategic and operational level.

The report should not exceed 10 pages and be submitted in a PDF format.
Mention your group number and the names and student ids of all team members on the cover page of your report.

DELIVERABLES

2/4: Solution file

Your proposed solution should be entered into a solution spreadsheet (**template provided on Ufora - PPP_solutionFile_2026.xlsx**).
The solution itself should include the following:

1. The number and constitution (= the SKUs included) of the packs that you want to use. Each record (row) represents a pack, and the columns refer to SKU identifiers. The values in the table represent the number of units of a specific SKU that are allocated to a pack.

2. The allocation of packs to each sales channel. The records (rows) again represent the identifiers for the packs, identical to the identification in the first tab. The columns represent the sales channels. The numbers in this table represent the number of packs that are allocated to a specific channel.

Do not add any other tabs, calculations, or macros to this spreadsheet. This should only be used to hand in your solution, nothing more.

SOLUTION FILE

Packs tab: How your packs are constituted

Enter an integer number >= 0, indicating how many units of an SKU are in a pack.

The identity matrix that you start with indicates one SKU per pack.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	A
3		FBE5E6Q4S6_XS	FBE5E6Q4S6_S	FBE5E6Q4S6_M	FBE5E6Q4S6_L	FBE5E6Q4S6_XL	FBE5E6Q4S6_XXL	B5K8O3NA22_XS	B5K8O3NA22_S	B5K8O3NA22_M	B5K8O3NA22_L	B5K8O3NA22_XL	B5K8O3NA22_XXL	I8RO5VPQIV_XS	I8RO5VPQIV_S	I8RO5VPQIV_M	I8RO5VPQIV_L	I8RO5VPQIV_XL	I8RO5VPQIV_XXL	SUSBYBGRMB_XS	SUSBYBGRMB_S	SUSBYBGRMB_M	SUSBYBGRMB_L	SUSBYBGRMB_XL	SUSBYBGRMB_XXL	DY2XGBW1QT_XS	DY2XGBW1QT_S	DY2XGBW1QT_M	DY2XGBW1QT_L	DY2XGBW1QT_XL	DY2XGBW1QT_XXL	EC1T9WA177_XS	EC1T9WA177_S	EC1T9WA177_M	EC1T9WA177_L	EC1T9WA177_XL
4	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	3	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	4	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	5	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	6	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
11	7	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	8	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	9	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	10	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15	11	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	12	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
17	13	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
19	15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
20	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
21	17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
22	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23	19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
25	21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
26	22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
27	23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
28	24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0

Checks | Assortment | Packs | PackAllocation | +

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SOLUTION FILE

PackAllocation tab: How many packs are to be sent where?

Rows correspond to the packs as they have been defined.

Again integer numbers >= 0 can added to make up your solution

	A	B	C	D	E	F	G	H	I	J	K	L	M	Q	R	S
1		Stockholm	Copenhagen	Helsinki	Paris	Berlin	Madrid	Rome	Amsterdam	Brussels	Webshop	Platform				
2	0	5	1	0	8	1	8	0	8	2	9	9				
3	1	6	5	10	10	4	1	6	1	4	3	7				
4	2	0	0	0	4	0	0	1	0	0	0	0				
5	3	1	5	8	6	10	9	10	2	9	7	7				
6	4	5	7	1	9	10	0	9	5	4	10	5				
7	5	6	0	0	9	5	1	10	5	7	4	0				
8	6	10	2	10	8	10	1	4	7	3	1	6				
9	7	5	7	8	8	9	8	10	0	8	6	2				
10	8	9	4	2	8	8	3	8	1	6	9	6				
11	9	1	8	8	7	8	6	10	0	1	5	6				
12	10	9	1	0	2	1	0	8	0	4	7	1				
13	11	2	8	2	5	3	5	9	2	5	9	10				
14	12	0	3	1	7	1	3	4	6	9	10	2				
15	13	6	10	9	4	7	2	5	3	8	7	4				
16	14	8	3	1	3	9	8	1	5	8	9	4				
17	15	10	4	4	6	7	6	9	0	1	9	10				
18	16	0	3	7	6	5	2	6	2	4	0	6				
19	17	0	10	10	3	9	8	8	2	2	10	8				
20	18	8	5	3	4	6	9	8	7	5	9	9				
21	19	1	9	5	4	3	6	6	1	6	8	8				
22	20	6	4	9	0	2	9	6	4	0	2	1				
23	21	1	8	8	4	1	0	10	10	4	10	0				
24	22	7	6	3	10	2	5	1	6	2	5	1				
25	23	10	1	3	9	1	10	4	5	9	1	6				
26	24	3	4	10	6	3	4	9	8	7	4	9				
27	25	0	10	0	5	3	8	8	7	2	4	6				
28	26	6	4	0	4	7	5	10	5	2	4	9				
29	27	4	3	0	7	3	10	10	1	8	3	2				
30	28	2	3	0	5	2	10	7	4	3	6	4				
31	29	0	4	6	4	4	9	8	3	3	3	4				
32	30	5	7	6	5	6	2	0	3	2	1	8				

Checks | Assortment | Packs | PackAllocation

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SOLUTION FILE

Simple checks: Assortment

Shows how many SKU end up in a specific location, and shows totals. Use this to check if your solution matches with this before you submit.

Assortment resulting from solution

SKU	Stockholm	Copenhagen	Helsinki	Paris	Berlin	Madrid	Rome	Amsterdam	Brussels	Webshop	Platform	SKU Totals
DVAZLH292T_XS	6	4	8	3	3	4	10	6	9	9	2	64
DVAZLH292T_S	10	9	8	8	9	6	9	3	6	10	6	84
DVAZLH292T_M	8	0	8	6	7	4	5	5	6	1	3	53
DVAZLH292T_L	7	1	6	6	3	2	3	9	4	9	10	60
DVAZLH292T_XL	3	4	0	9	0	8	3	9	4	4	3	47
UKDVICMCMI_XXS	10	2	3	9	10	7	9	0	2	5	2	59
UKDVICMCMI_XS	8	4	6	1	8	10	8	3	10	6	2	66
UKDVICMCMI_S	6	0	5	3	2	8	7	1	8	2	8	50
UKDVICMCMI_M	1	7	5	8	7	9	3	2	9	3	5	59
UKDVICMCMI_L	8	0	5	0	10	3	5	9	3	8	5	56
UKDVICMCMI_XL	7	1	4	9	5	5	10	10	7	2	4	64
F58Y8I60TY_XXS	6	5	6	2	0	2	9	7	4	6	4	51
F58Y8I60TY_XS	5	2	3	8	9	9	9	10	9	1	0	65
F58Y8I60TY_S	5	5	8	8	3	6	4	6	3	8	3	59
F58Y8I60TY_M	8	10	3	4	10	7	3	2	0	4	2	53
F58Y8I60TY_L	8	1	7	3	2	8	7	6	9	3	1	55
F58Y8I60TY_XL	4	9	8	2	3	10	8	9	3	0	1	57
ARCCMK4UZC_onesize	1	4	2	2	5	0	6	7	0	6	10	43
J7DERD76U8_onesize	8	4	0	5	4	10	0	10	6	8	3	58
B5ZJF102L8_onesize	0	0	0	3	7	9	10	5	5	10	7	56
KBG0WAZV9P_onesize	0	4	2	10	8	8	1	5	9	4	9	60
KBX5F1FCPW_onesize	0	6	8	1	2	2	7	5	7	4	0	42
Location totals	1343	1260	1333	1469	1400	1438	1344	1362	1350	1384	1410	

Checks | Assortment | Packs | PackAllocation

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SOLUTION FILE

Simple checks: Assortment

Shows how many SKU end up in a specific location, and shows totals. Use this to check if your solution matches with this before you submit.

Assortment resulting from solution

SKU	Stockholm	Copenhagen	Helsinki	Paris	Berlin	Madrid	Rome	Amsterdam	Brussels	Webshop	Platform	SKU Totals
DVAZLH292T_XS	6	4	8	3	3	4	10	6	9	9	2	64
DVAZLH292T_S	10	9	8	8	9	6	9	3	6	10	6	84
DVAZLH292T_M	8	0	8	6	7	4	5	5	6	1	3	53
DVAZLH292T_L	7	1	6	6	3	2	3	9	4	9	10	60
DVAZLH292T_XL	3	4	0	9	0	8	3	9	4	4	3	47
UKDVICMCMI_XXS	10	2	3	9	10	7	9	0	2	5	2	59
UKDVICMCMI_XS	8	4	6	1	8	10	8	3	10	6	2	66
UKDVICMCMI_S	6	0	5	3	2	8	7	1	8	2	8	50
UKDVICMCMI_M	1	7	5	8	7	9	3	2	9	3	5	59
UKDVICMCMI_L	8	0	5	0	10	3	5	9	3	8	5	56
UKDVICMCMI_XL	7	1	4	9	5	5	10	10	7	2	4	64
F58Y8I60TY_XXS	6	5	6	2	0	2	9	7	4	6	4	51
F58Y8I60TY_XS	5	2	3	8	9	9	9	10	9	1	0	65
F58Y8I60TY_S	5	5	8	8	3	6	4	6	3	8	3	59
F58Y8I60TY_M	8	10	3	4	10	7	3	2	0	4	2	53
F58Y8I60TY_L	8	1	7	3	2	8	7	6	9	3	1	55
F58Y8I60TY_XL	4	9	8	2	3	10	8	9	3	0	1	57
ARCCMK4UZC_onesize	1	4	2	2	5	0	6	7	0	6	10	43
J7DERD76U8_onesize	8	4	0	5	4	10	0	10	6	8	3	58
B5ZJF102L8_onesize	0	0	0	3	7	9	10	5	5	10	7	56
KBG0WAZV9P_onesize	0	4	2	10	8	8	1	5	9	4	9	60
KBX5F1FCPW_onesize	0	6	8	1	2	2	7	5	7	4	0	42
Location totals	1343	1260	1333	1469	1400	1438	1344	1362	1350	1384	1410	

Checks | Assortment | Packs | PackAllocation

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SOLUTION FILE

Simple checks: Checks

Some very basic sanity checks. If these do not check out you have probably made a mistake.

Note: This is not sufficient to guarantee that you have not made mistakes!

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	**Solution checkcells**												
2													
3	**Checkcells**												
4													
5		275	Total SKU included										
6													
7	**Basic statistics**												
8													
9		275	Total number of packs defined										
10		275	Total number of SKU that are present in at least 1 pack										
11		275	Tota number of unique packs used in allocation										
12		15093	Total number of packs used in allocation										
13													
14	**Errors and warning**												
15													
16		0	=	0	TRUE	The number of defined packs should be equal to the packs used (in the final solution)							
17		0	=	0	TRUE	The number of products not allocated to a pack should be zero							
18													

Checks | Assortment | Packs | PackAllocation | +

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DELIVERABLES

3/4: Your code or calculations

Also upload any **python code or spreadsheets** that you have used to create your solution. If possible, think about the legibility of this code for an outside person. Generally, this will be used to evaluate the technical quality of your approach if you have made a big mistake along the way (and therefore score low on the automatic technical evaluation of your solution quality).

DELIVERABLES

4/4: Presentation

A 5-minute elevator pitch, focus on:

- Your methodology, why have you used this?
- Important assumptions that you made, that impacted your solution
- The value of your solution to sustainable essentials
- Pitfalls, risks, extensions, limitations,...

This is followed by 10 minutes of questions, focusing on things that were not clear in your report or presentation.

EVALUATION CRITERIA (40% OF COURSE TOTAL)

Criterion	Description	Weight
Strategic Insights	Validity of the strategic analysis and suggested managerial decisions.	10%
Forecasting Methodology	Justification of chosen forecasting methods. Comparison based on accuracy, simplicity, scalability, and computational costs.	20%
Optimization Methodology	Clarity in criteria for optimization. Use of sensitivity analysis. Appropriateness of the methodology for the problem.	20%
Quantitative Results	The performance of your final solution, as provided in your solution files. Performance will be evaluated based on the best minimization of all associated costs.	30%
Presentation and Questions	Clarity, engagement, and professionalism of the presentation. Effective use of storytelling and visual aids. Quality of answers to follow-up questions on the case study.	20%

[UPDATED] CLASS SCHEDULE

date	start	Topic	Assignment
12/02/2026	13:00	Supply chain strategy	
19/02/2026	13:00	Network design	
26/02/2026	14:00	Forecasting in supply chains	
5/03/2026	13:00	Deterministic inventory management	**Briefing group assignment**
12/03/2026	13:00	Stochastic inventory management	
19/03/2026	13:00	No class	**Office hours on 18th March**
26/03/2026	13:00	No class	
2/04/2026	13:00	Transportation in supply chains	**Office hours (before and after class)**
9/04/2026	-	No class - easter holidays	
16/04/2026	-	No class - easter holidays	
23/04/2026	13:00	Price and revenue management	**Office hours in the morning (before class)**
30/04/2026	13:00	Sustainable SCM + Guest lecture: Multi-echelon safety stock	**Office hours in the morning (before class)**
7/05/2026	13:00	Presentations for group assignments	

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KEY DATES AND DEADLINES

- Office hours for Q&A on the group assignment on the following days.
  - Wednesday 18/3
  - Thursday 2/4
  - Thursday 23/4
  - Thursday 30/4
- Meeting slots will be opened up on Ufora to make appointments.
- These take place at my office (2nd floor campus Tweekerken)
- **All files must be submitted by the 5th of May at 23:59**
- Presentations take place on the 7th of May

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